

electron is much larger than protons.

27 B

$$\text{Electrons: } dN_e/dt = I / e = 2.0 \times 10^{-6} / 1.6 \times 10^{-19}$$

$$\text{Photons: } dN_p/dt = P / E = 0.31 \times 10^{-3} / 3.11 \times 1.6 \times 10^{-19}$$

$$\text{Ratio} = 0.020$$

28 C

$$p = mv = (0.11 \times 10^{-31})(1.50 \times 10^6) = 1.67 \times 10^{-24} \text{ kg m s}^{-1}$$